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# A paleomagnetic investigation of rocks from the Ohio Range and the Dry Valleys, Transantarctic Mountains, Antarctica

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**Abstract** Two well-defined virtual geomagnetic poles (VGPs) for East Antarctica were obtained from the Middle Jurassic Ferrar Dolerite, sampled from a thick sill on Mt Schopf in the Ohio Range, and from a horizontal sheet intruding Paleozoic granitic rocks at Mt Cerberus in the Dry Valleys. The VGP from the sill at Mt Schopf lies at lat. 58°.0S, long. 129°.0W (dm=13°, dp=12°), and the VGP from Mt Cerberus lies at lat. 57°.8S, long. 135°.7W (dm=6°, dp=6°). Rocks from both localities have normal polarity magnetisations. These data, together with 15 other acceptable pole positions reported in the literature, define a Middle Jurassic paleomagnetic pole at 52°.7S, 139°.6W ($\alpha_{95}$=4°.4), which is not significantly different from previously reported mean Middle Jurassic paleomagnetic poles from East Antarctica.

Paleomagnetic investigations in the Ohio Range of mafic inclusions in Ordovician granitic and sedimentary rocks of Devonian and Permian ages found that the rocks are magnetically unstable and are unreliable for determining a paleomagnetic pole.

**Keywords** paleomagnetism; Antarctica; Ohio Range; Dry Valleys; Ferrar Dolerite; Middle Jurassic

## INTRODUCTION

Paleomagnetic pole positions determined from Paleozoic rocks of the East Antarctic craton are extremely sparse; only poles for Ordovician rocks have been determined with any certainty (Zijderveld 1968; Manzoni & Nanni 1977). By contrast, the Jurassic pole for East Antarctica is well established from paleomagnetic studies (Bull & Irving 1960; Bull et al. 1962; Blundell 1966; Ostrander 1971; Beck 1972; Funaki 1984) of intrusive rocks from the Jurassic Ferrar Dolerite (Kyle et al. 1981). The present study was started during an expedition to the Ohio Range, southern Transantarctic Mountains (Fig. 1), sponsored by the New Zealand Antarctic Research Programme during the 1979–80 field season, and was continued on a visit to the Dry Valleys (Fig. 1) sponsored by the United States Antarctic Research Program during the 1984–85 field season. One objective of the study was to collect samples for paleomagnetic analysis, of Ordovician granitic rocks and Devonian and Permian sedimentary rocks from the Ohio Range, to increase our understanding of the Paleozoic apparent polar wander path for East Antarctica. Another objective was to augment previous determinations of the Middle Jurassic paleomagnetic pole with additional virtual paleomagnetic poles (VGPs) from the Middle Jurassic Ferrar Dolerite (Kyle et al. 1981).

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## GEOLOGIC SETTING

### Ohio Range

The Ohio Range comprises a prominent 1000 m granitic escarpment separating the Polar Plateau to the south from the great ice sheet of Marie Byrd Land to the north (Fig. 1). Ordovician granitic basement rocks (Treves 1965) compose most of the steep escarpment, and are overlain by a mostly flat lying, 1200 m thick sequence of nonmetamorphosed sedimentary rocks of Devonian and Permian age (Long 1965; Bradshaw et al. 1984; McCartan et al. 1986). A prominent basaltic dolerite sill, correlated with the Ferrar Dolerite, intrudes the upper part of the sedimentary sequence (Treves 1965). The basement rocks of the Ohio Range are part of the Wisconsin Range batholith (Mirsky 1969), inferred to have been emplaced during the Late Cambrian–Ordovician Ross Orogeny (Gunn & Warren 1962). These granitic rocks in the Ohio Range consist of biotite quartz monzonite and

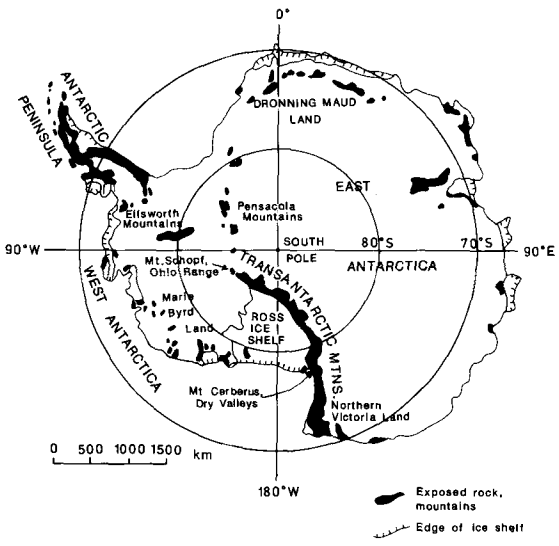



**Fig. 1** Antarctica, showing the Transantarctic Mountains and the positions of Mt Schopf, Ohio Range and Mt Cerberus, Dry Valleys.

biotite granodiorite (Treves 1965). The biotite quartz monzonite locally cuts the biotite granodiorite, but in other places the contact is diffuse, which suggests that the two units are approximately coeval. Both rock types are coarse grained, slightly gneissic and porphyritic, and contain microcline phenocrysts as long as 20 mm.

Treves (1965) reported K-Ar and Rb-Sr ages from the biotite quartz monzonite from Discovery Ridge and Treves Butte in the Ohio Range — a biotite K-Ar age of 472±24 Ma, a feldspar (model?) Rb-Sr age of 471±49 Ma, and a whole rocks (model?) Rb-Sr age of 516±72 Ma. From these data it is inferred that the basement granitic rocks of the Ohio Range are about Middle Ordovician in age.

Numerous dioritic, diabasic, and metavolcanic(?) and metasedimentary(?) inclusions, ranging from <10 mm to several tens of metres, are common in both the granodiorite and quartz monzonite; the granodiorite locally contains as much as 10% inclusions. Treves (1965) described the mineralogy of these inclusions in detail.

The layered sedimentary rocks overlying the basement consist, from bottom to top, of a 0–50 m thick, largely marine sequence — the Lower Devonian Horlick Formation (Bradshaw & McCartan 1983; McCartan et al. 1986) — overlain by a c. 300 m thick sequence of grey to black, Permian, glacial deposits of Buckeye Tillite (Long 1965), then by c. 150 m of well-indurated shale of the Discovery Ridge Formation, and are capped by at least 700 m of coal-bearing sandstone and shale of the Upper Permian Mt Glossopteris Formation (Long 1965).

On Mt Schopf, a dolerite sill at least 180 m thick is intruded into the flat-lying Mt Glossopteris Formation. The sill has a tholeiitic basalt composition (Treves 1965), and the lower contact is chilled against hornfelsed shales of the Mt Glossopteris Formation; the upper contact is eroded.

The sill on Mt Schopf is undated but is clearly related to the sills and dikes of the Ferrar Dolerite that are common throughout the Transantarctic Mountains (Treves 1965). The Ferrar Dolerite that cuts Paleozoic Beacon Supergroup rocks in the Pensacola Mountains has been dated by K-Ar analysis on plagioclase at 179±5 Ma (Middle Jurassic) by Ford & Kistler (1980), who also report an age of 172±4 Ma on the Dufek layered intrusion. In the Transantarctic Mountains, $^{40}Ar/^{39}Ar$ age determinations on six intrusive rocks from the Ferrar Dolerite and one basalt flow from the Kirkpatrick Basalt give a mean age of 179±7 Ma (Kyle et al. 1981), although Kyle et al. (1981) suggest that a younger magmatic event at 165±2 Ma may also have occurred. An $^{40}Ar/^{39}Ar$ age of 174±10 Ma on the Kirkpatrick Basalt Group from northern Victoria Land (McIntosh et al. 1982) and a K-Ar age of 180±5 Ma on the Kirkpatrick Basalt from the central Transantarctic Mountains (Elliot et al. 1985) are also in excellent agreement with the age determination of Kyle et al. (1981).

## Dry Valleys

The Dry Valleys lie in the Royal Society Range of the Transantarctic Mountains, c. 1300 km northwest of the Ohio Range (Fig. 1). The oldest units in the Dry Valleys region consist of metasedimentary rocks of the Skelton Group (Gunn & Warren 1962), of Cambrian or late Proterozoic age, that are intruded by a suite of locally deformed granitic to dioritic rocks, called the Granite Harbour Intrusive Suite (Gunn & Warren 1962), which were emplaced during the Ordovician Ross Orogeny. Late synorogenic (Ordovician) lamprophyre dikes are locally common in the older rocks.

A widespread unconformity separates the lower Paleozoic rocks from the Permian Beacon Supergroup, and all units are pervasively intruded by sills and dikes of the Ferrar Dolerite.

## PALEOMAGNETIC METHOD

In the Ohio Range, all samples were collected as oriented blocks with a hand-held level and sun compass. Where cloud cover or shade prevented the use of a sun compass, magnetic declinations were measured. The difference between magnetic and sun declinations, where checked, was never more than about 2°. Samples in the Dry Valleys were collected with a gasoline-powered, hand-held drill, using ethylene glycol as drilling coolant, and were oriented with a sun compass.

In the laboratory, a 250 mm diameter oriented core was drilled, wherever possible, from each of the block samples from the Ohio Range. Due to the fissile nature of many of the shale and tillite samples, drilling was virtually impossible without first impregnating the samples with epoxy resin.

Remanent directions and intensities were measured on a 10 Hz spinner magnetometer, and demagnetisation techniques follow essentially those discussed by McElhinny (1973). Systematic alternating field (a.f.) demagnetisation was performed on all prepared samples; thermal demagnetisation was performed on representative samples for comparative purposes.

A description of the specific units sampled and the treatment undertaken follows.

### Ohio Range

Most paleomagnetic sampling was carried out on the older sedimentary units and the underlying granitic complex, because the Ohio Range Expedition was primarily concerned with stratigraphic and paleontologic studies of the Horlick Formation. A total of 95 oriented block samples was collected, from fine-grained dioritic inclusions in the basement granitic rock (29 samples), shale and sandstone layers in the Horlick Formation (36 samples), tillite and interbedded black shale from the Buckeye Tillite (22 samples), and dolerite from the Ferrar Dolerite on Mt Schopf (8 samples). Because most sedimentary beds are horizontally bedded, no structural corrections were applied. The dolerite sill on Mt Schopf was only briefly visited, and the samples were collected within 3.2 m of the lower contact with hornfelsed shales of the Mt Glossopteris Formation.

Neither the rocks from the Horlick Formation nor from the Buckeye Tillite are noticeably hematitic in hand specimen; their colour varies from pale grey to olive to dark grey. Considerable difficulty was encountered in collecting most samples due to the highly laminated, fissile nature of the rock, particularly in the Horlick Formation. Fine-grained dioritic inclusions in the basement granitic rock were suspected to be more magnetically stable than the enclosing rock. This suspicion followed from successful paleomagnetic studies of Cretaceous plutonic rocks from the Antarctic Peninsula (Kellogg & Reynolds 1978; Kellogg 1980), where inclusions in generally magnetically unstable granitic rock yielded well-defined, stable groupings of remanent direction. The magnetic stability of the inclusions was inferred to be due to the presence of small magnetite grains, each containing a small number of magnetic domains, as compared to the magnetite in the coarser grained granitic rock.

Alternating field (a.f.) demagnetisation was performed on 29 samples from the dioritic inclusions and on the 8 samples of dolerite from the Ferrar, up to a peak field of 70 milliteslas (mT). After the measurement of the natural remanent magnetisation (NRM), demagnetisation steps were 5, 10, 20, 30, 40, 50, and 70 mT. Six pilot samples from the inclusions and two pilot samples of the dolerite were also subjected to step-wise thermal demagnetisation up to 580°C. A vector-projection diagram, similar to that described by Zijderveld (1967), shows the results of a.f. demagnetisation of a representative sample of dolerite from Mt Schopf (Fig. 2A).

To test the stability of magnetisation in the sedimentary rocks, four samples each from the Horlick Formation and the Buckeye Tillite were chosen for step-wise a.f. demagnetisation up to a peak field of 70 mT. Three samples for which impregnation with epoxy was not necessary also were selected from each unit for step-wise thermal demagnetisation.

### Dry Valleys

Eleven samples were collected from one locality in Ferrar Dolerite on Mt Cerberus. The intrusion is >100 m thick, and the samples were drilled about 30 m above the horizontal contact with underlying granite of the Granite Harbour Intrusive Suite. In the laboratory, pilot samples were cleaned by a.f. demagnetisation to 70 mT and by thermal demagnetisation to 600°C (Fig. 2B, and 2C). Remaining samples were then cleaned in steps up to 50 mT.

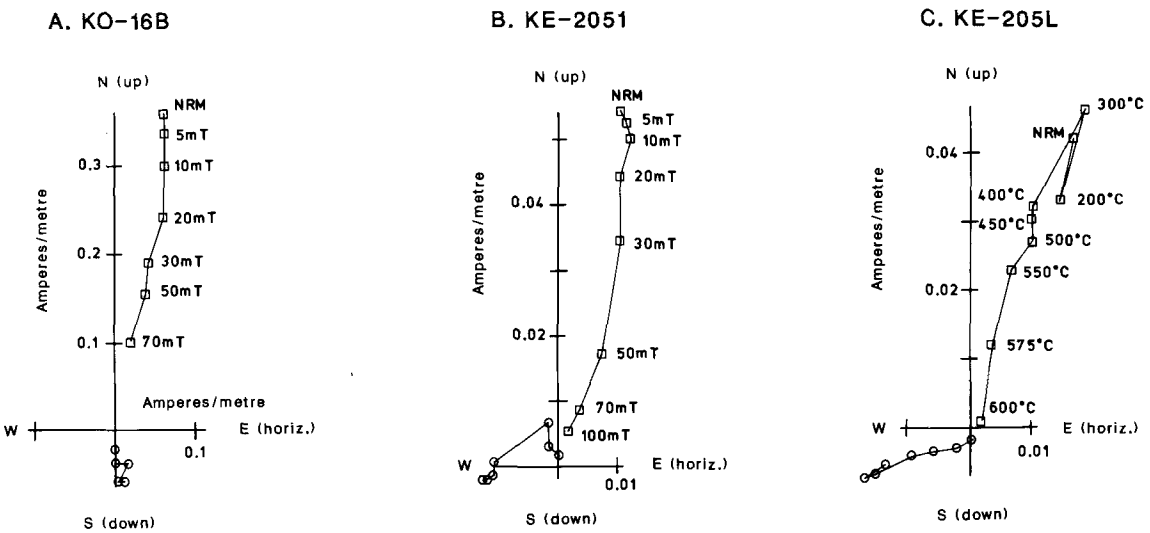



**Fig. 2** Vector demagnetisation plots for samples of Ferrar Dolerite from Mt Schopf, Ohio Range and Mt Cerberus, Dry Valleys. Circles represent the projection of the total vector onto the horizontal plane (N and E axes); squares represent the total vector as shown on a vertical plane (up and horizontal axes). Both axes in amperes/metre. **A** a.f. demagnetisation to 70 mT for representative sample from Mt Schopf. **B** a.f. demagnetisation for representative sample from Mt Cerberus. **C** Thermal demagnetisation to 600°C for representative sample from Mt Cerberus.

## PALEOMAGNETIC RESULTS

### Ferrar Dolerite

*Mt Schopf, Ohio Range*

Although only eight samples of Ferrar Dolerite on Mt Schopf were sampled, the remanent directions are extremely stable, single component (Fig. 2A), and define a well-grouped cluster of directions (Fig. 3A). After demagnetisation to 10 mT, the mean direction did not vary more than 4°, and the semi-angle of the cone of 95% confidence ($\alpha_{95}$) remained at about 7°. NRM intensities varied between $4 \times 10^{-2}$ and $5 \times 10^{-1}$ A/m, and after 50 mT demagnetisation the intensities were between about 0.1 and 0.6 of the NRM values.

The mean direction after 50 mT a.f. demagnetisation is D = 163°.0, I = –75°.7, with $\alpha_{95}$ = 7°.2 (k = 60.3). This direction defines a virtual geomagnetic pole (VGP) at latitude 58°.0S, longitude 129°.0W (Table 1).

*Mt Cerberus, Dry Valleys*

Of the 11 samples of dolerite collected on Mt Cerberus, 9 clustered closely at all demagnetisation levels (Fig. 3B); 2 samples clearly have highly divergent directions and are therefore rejected from the calculation of the mean direction. The 9 clustered samples have a mean direction of D = 263°.0, I = –74°.8, with $\alpha_{95}$ = 3°.3 (k = 245.2). This direction defines a VGP at latitude 57°.8S, longitude 135°.7W (Table 1).

The results of a.f. and thermal demagnetisation (Fig. 2B, and 2C) showed that the magnetisation was single component. Between thermal demagnetisation steps of 575°C and 600°C, remanent intensities drop to zero, indicating that the blocking temperature is near 580°C, the Curie temperature for pure magnetite.

### Inclusions in granite

The results of both a.f. and thermal demagnetisation of the fine-grained dioritic inclusions in granitic rock from the Ohio Range were disappointing. NRM directions showed a large viscous component, having steep negative inclinations parallel to the Earth's present field (Fig. 4A). NRM intensities were typically $1–3 \times 10^{-2}$ A/m. A.f. demagnetisation to 10 mT removed the viscous component, including about 60% of the NRM intensity. Such a rapid drop in intensity suggests that most magnetisation resides in large multidomain grains of magnetite. For most samples, the directions differed greatly between successively higher levels of demagnetisation, indicating unstable behaviour. Intensities after demagnetisation to 50 mT (Fig. 4B) were typically

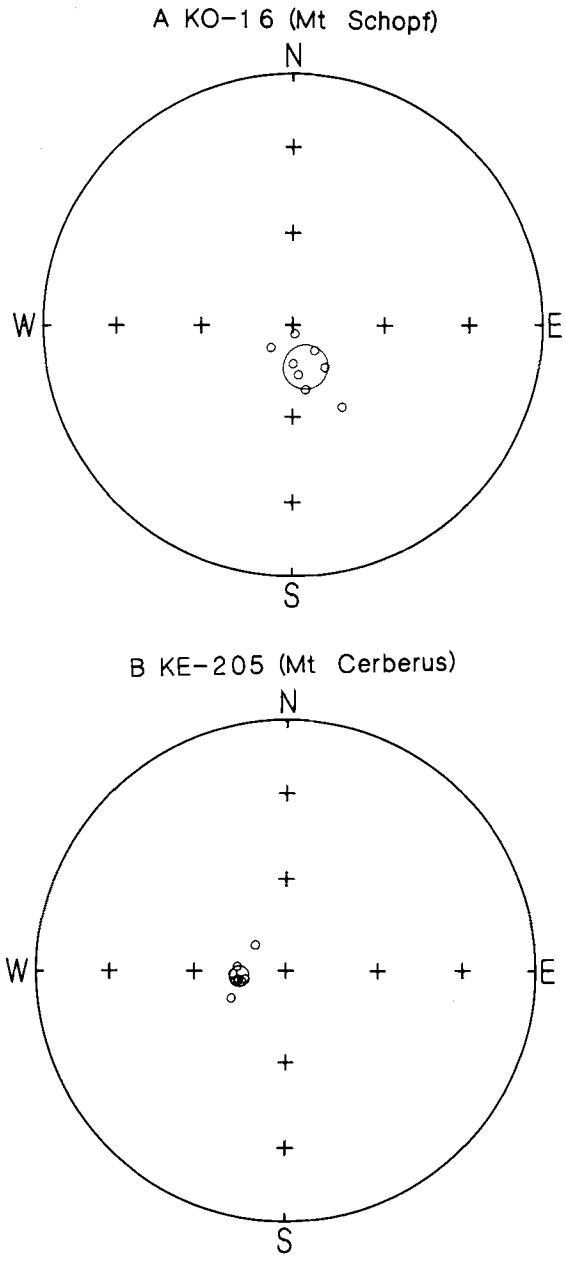



**Fig. 3** Equal-area projections of a.f.-cleaned samples of Ferrar Dolerite from Mt Schopf (A) and Mt Cerberus (B), demagnetised to 50 mT and 30 mT, respectively. The cone of 95% confidence is indicated on each diagram.

20% of NRM intensity. Although there is a clustering of down-to-the-southeast directions after 50 mT demagnetisation (most of these directions are from some inclusions, about 1–2 m in diameter, sampled on Treves Butte in the northwest part of the Ohio Range), most directions are widely scattered, and the results are therefore considered unreliable. The six samples subjected to thermal demagnetisation likewise demonstrated unstable behaviour.

### Horlick Formation and Buckeye Tillite

The pilot samples from the Horlick Formation and Buckeye Tillite contained low intensities of magnetisation and exhibited extremely unstable behaviour during both a.f. and thermal demagnetisation. Directions wandered widely from one demagnetisation level to the next, and magnetisation intensities, after initially decreasing, fluctuated both up and down. NRM intensities were variable but ranged between $10^{-3}$ and $10^{-4}$ A/m, approaching the detection limit for the spinner magnetometer. The results of both a.f. demagnetisation to 70 mT and thermal demagnetisation to 450°C (the maximum temperature for which a remanence could be measured for most samples) demonstrated that it was pointless to prepare and measure additional samples.

## DISCUSSION

Little information regarding a Paleozoic paleomagnetic pole for East Antarctica can be discerned from the study of Paleozoic rocks from the Ohio Range. The sampled dioritic inclusions in the Ordovician basement granites proved to be magnetically unstable. Similarly, pilot samples of shale and tillite from the overlying Devonian and Permian sedimentary rocks proved very weak and magnetically unstable. Problems in preparing samples of fissile shale suitable for thermal demagnetisation also precluded intensive study. Even if well-defined paleomagnetic directions had been measured in the Paleozoic sedimentary rocks, the possible effects of remagnetisation by the large dolerite sill on Mt Schopf, stratigraphically only c. 700 m above some of the sampled localities, might bring into question the reliability of the results; such arguments were used by Beck (1972) to reject paleomagnetic results from Permian Beacon Supergroup sedimentary rocks from elsewhere in the Transantarctic Mountains.

In contrast to the Paleozoic rocks, the Middle Jurassic Ferrar Dolerite intrusives on Mt Schopf and Mt Cerberus provided well-defined VGPs (Table 1). A well-defined Middle Jurassic pole is critical in plate reconstructions between East Antarctica and

**Table 1** Summary of paleomagnetic data for Ferrar Dolerite from Mt Schopf, Ohio Range, and Mt Cerberus, Dry Valleys, Antarctica. Only data for NRM and demagnetisation level (Demag level) chosen to represent the locality are shown. The coordinates of the collecting locality on Mt Schopf are 84°.80S, 113°.58W; the coordinates at Mt Cerberus are 77°.43S, 161°.75E.

| Demag | | | | | | | Virtual geomagnetic pole | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| level (mT) | N | I | D | R | k | $\alpha_{95}$ | Lat. | Long. | dm | dp |
| *Mt Schopf, Ohio Range* | | | | | | | | | | |
| NRM | 8 | –80.1 | 177.1 | 7.681 | 21.9 | 12.1 | – | – | – | – |
| 50 | 8 | –75.7 | 163.0 | 7.884 | 60.3 | 7.2 | 58.0S | 129.0W | 13 | 12 |
| *Mt Cerberus, Dry Valleys* | | | | | | | | | | |
| NRM | 9 | –76.6 | 264.2 | 8.948 | 154 | 4.2 | – | – | – | – |
| 30 | 9 | –74.8 | 263.2 | 8.967 | 245 | 3.3 | 57.8S | 135.7W | 6 | 6 |

N is the number of samples; I is mean inclination, negative upward; D is mean declination, clockwise from true north; R is the magnitude of the vector sum of the N unit vectors; k is Fisher's (1953) best estimate of precision; $\alpha_{95}$ is the semi-angle of the cone of 95% confidence; dm and dp are the semi-major and semi-minor axes, respectively, of the oval of 95% confidence.

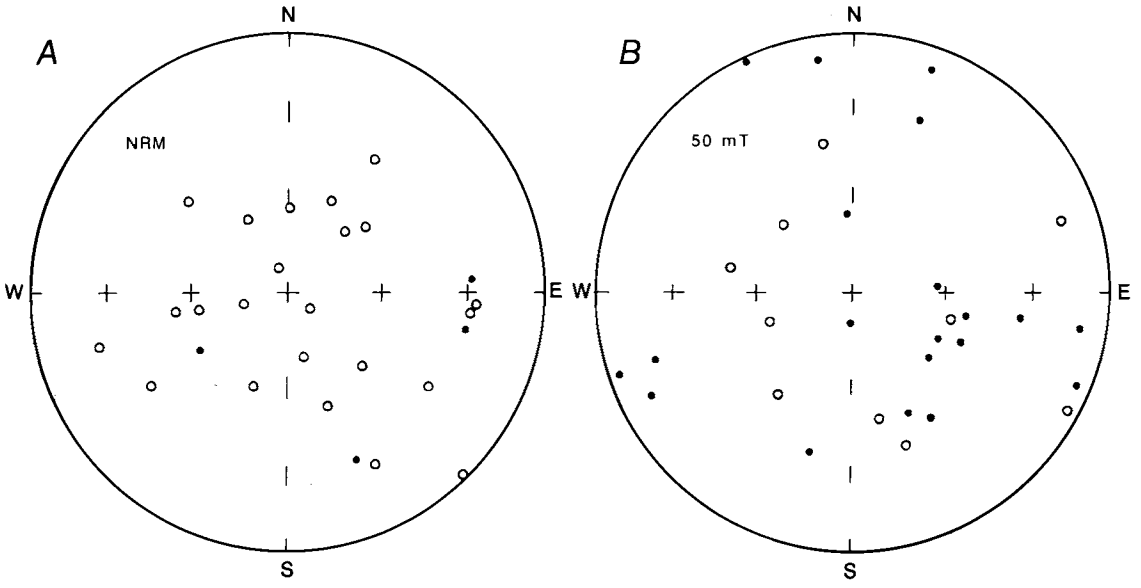



**Fig. 4** Equal-area plots of paleomagnetic direction from mafic inclusions in basement granite from the Ohio Range. Open circles are upper hemisphere; closed circles are lower hemisphere. **A** NRM (natural remanent magnetisation) directions. **B** Directions after 50 mT demagnetisation.

the various crustal fragments that may make up West Antarctica (Dalziel et al. 1987), and several attempts have been made to calculate such a pole (Funaki 1984; Grunow et al. 1987). Paleomagnetic data, for example, indicate that the Antarctic Peninsula and the Ellsworth Mountains may have been part of a single crustal block since Middle Jurassic time, and that 16° of counterclockwise movement restores the Antarctic Peninsula/Ellsworth Mountains paleomagnetic pole with that determined from previous studies of the Ferrar Dolerite and rocks of equivalent age (Grunow et al. 1987).

To better define the paleomagnetic pole for the Ferrar, the results from this study were combined with a number of other studies from the literature (Table 2 and Fig. 5). An attempt was made to subdivide each study into the results from a single sampling locality or group of clustered localities.

**Table 2** Paleomagnetic data used in calculation of the Middle Jurassic paleomagnetic pole for East Antarctica. Samples from all localities have been cleaned to at least 15 mT.

| | $N_s$ | $N_c$ | $\alpha_{95}$ | k | Plat (°S) | Plong (°W) | No. on Fig. 5 | Reference |
|---|---|---|---|---|---|---|---|---|
| Ferrar Dolerite | | | | | | | | |
| Dry Valleys area | | | | | | | | |
| "Beacon intrusives" | 18 | – | 7 | – | 36.8 | 154.9 | 1 | Bull & others 1962 |
| "Peneplain sheet" | 24 | – | 5 | – | 44.1 | 145.1 | 2 | Bull & others 1962 |
| "Basement sheet" | 41 | – | 5 | – | 45.1 | 141.9 | 3 | Bull & others 1962 |
| Wright Valley | 26 | 1 | 2.4 | 136 | 45.3 | 152.0 | 4 | Funaki 1984 |
| Allan Hills | 19 | 3 | 5.1 | 44 | 47.0 | 133.2 | 5 | Funaki 1984 |
| Mt Fleming | 15 | 1 | 1.0 | 1466 | 68.6 | 139.5 | 6 | Funaki 1984 |
| Mt Cerberus | 9 | 1 | 3.3 | 245 | 57.8 | 135.7 | 7 | This study |
| Queen Alexandra Range | – | 7 | 5.4 | – | 54.2 | 139.8 | 8 | Ostrander 1971 |
| Beardmore Glacier | 13 | 9 | 11 | 18 | 59 | 139 | 9 | Briden & Oliver 1963 |
| Ohio Range | 8 | 1 | 7.2 | 60 | 58.0 | 129.0 | 10 | This study |
| Dufek layered intrusion | 91 | – | – | – | 60 | 137 | 11 | Beck 1972; Burmister & Sheriff 1980 |
| Mixed flows and dikes | | | | | | | | |
| Dronning Maud Land | 142 | 32 | 3.8 | 47 | 41.8 | 133.5 | 12 | Løvlie 1979 |
| Kirkpatrick Basalt | | | | | | | | |
| Queen Alexandra Range | | | | | | | | |
| Storm Peak | – | 12 | 6.7 | – | 44.1 | 128.5 | 13 | Ostrander 1971 |
| Mt Falla | – | 14 | 4.4 | – | 53.8 | 137.4 | 14 | Ostrander 1971 |
| Mesa Range | 60 | 15 | 1 | 174 | 64 | 150 | 15 | McIntosh & others 1982 |
| David Glacier area | | | | | | | | |
| Brimstone Peak | 29 | 12 | 4 | 156 | 56 | 142 | 16 | McIntosh & others 1982 |
| Gorgon Peak | 26 | 13 | 4 | 122 | 56 | 130 | 17 | McIntosh & others 1982 |

$N_s$ = number of samples; $N_c$ = number of sites; $\alpha_{95}$ = half angle of the cone of 95% confidence, in degrees; K = estimate of precision parameter (Fisher 1953); and Plat and Plong are the paleolatitude and paleolongitude, respectively, of the calculated virtual geomagnetic pole.

The following criteria were adopted in selecting data: (1) at least eight samples must have been collected at each locality or clustered group of localities; (2) the samples must have been a.f. demagnetised to at least 15 mT; and (3) the $\alpha_{95}$ at any one site must be smaller than 15°. Under these requirements, the early paleomagnetic results of Blundell & Stephenson (1959), and Turnbull (1959) on Ferrar rocks were rejected, as they did not employ systematic demagnetisation techniques. The paleomagnetic pole from the Dufek layered intrusion, calculated by Beck (1972), must also be rejected on the basis of studies by Ford et al. (1980), who determined that the layering was tilted and the temperature of the intrusion was above the blocking temperature of the carriers of magnetisation. Thus, a bedding correction applied by Beck (1972) is not appropriate, and the reinterpretation by Burmister & Sheriff (1980) is taken in Table 2 to supplant Beck's data. Data from Dronning Maud Land (Løvlie 1979) cannot be subdivided into results from different sites because tectonic corrections necessary for each site are not given; only the overall paleomagnetic pole is calculated.

When each of the 17 virtual geomagnetic poles and paleomagnetic poles in Table 2 is weighted equally, they define a mean paleomagnetic pole at 52°.7S, 139°.6W($\alpha_{95}$ = 4°.4), which is in excellent agreement with a mean paleomagnetic pole at 54°.8S, 139°.7W, ($\alpha_{95}$ = 3°.9) calculated from 15 localities in Ferrar rocks (many the same as for this study) along the Transantarctic Mountains (Grunow et al. 1987) and the mean pole at 53°.9S, 141°.8W ($\alpha_{95}$ = 6°.3) calculated from 10 localities by Funaki (1984).

The number of paleomagnetic studies, and the precision in dating the rocks, of the Ferrar Dolerite, make the Middle Jurassic paleomagnetic pole for East Antarctica one of the most precisely established in the world. However, there is still an almost complete lack of paleomagnetic poles for pre-Middle Jurassic rocks. This study points out a few of the pitfalls in trying to acquire reliable paleomagnetic information from these older rocks.

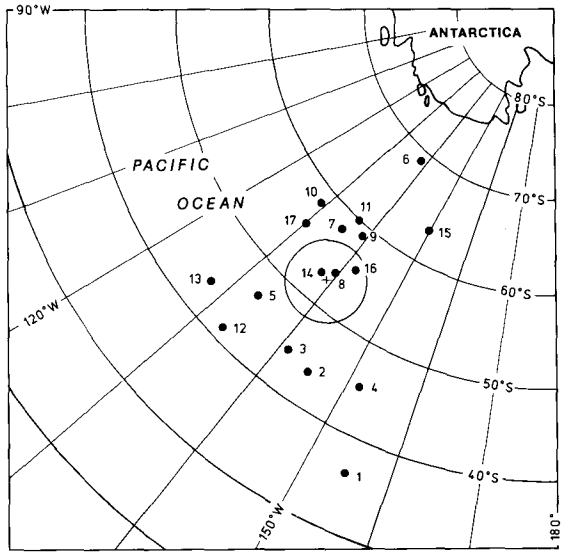



**Fig. 5** Equal-area projection of a portion the Earth's Southern Hemisphere, showing the virtual geomagnetic and paleomagnetic pole positions derived from this and other studies. Numbers refer to poles listed in Table 2. Oval of 95% confidence is drawn around mean pole position (+).

Further detailed paleomagnetic studies may yield reliable poles, but the problems of defining paleomagnetic results from sedimentary rocks, and overprinting by the widespread intrusion of the Ferrar Dolerite along almost the entire length of the Transantarctic Mountains, will make this a formidable task.

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## REFERENCES

Beck, M. E. 1972: Palaeomagnetism and magnetic polarity zones in the Jurassic Dufek intrusion, Pensacola Mountains, Antarctica. *Geophysical journal of the Royal Astronomical Society 28*: 49–63.

Blundell, D. J. 1966: Palaeomagnetism of the dolerite intrusions: Trans-Antarctic Expedition, 1955–58. *British Antarctic Survey scientific report 8*: 61–67.

Blundell, D. J.; Stephenson, P. J. 1959: Palaeomagnetism of some dolerite intrusions from the Theron Mountains and Whichaway Nunataks, Antarctica. *Nature 184*: 1860.

Bradshaw, M. A.; McCartan, L. 1983: The depositional environment of the Lower Devonian Horlick Formation, Ohio Range. Pp. 238–241 *in*: Oliver, R. L.; James, P. R.; Jago, J. B. *ed.* Antarctic earth science. Canberra, Australian Academy of Science. 697 p.

Bradshaw, M. A.; Newman, J.; Aitchison, J. C. 1984: Preliminary geological results of the 1983–84 Ohio Range expedition. *New Zealand Antarctic record 5 (3)*: 1–17.

Briden, J. C.; Oliver, R. L. 1963: Paleomagnetic results from the Beardmore Glacier region, Antarctica. *New Zealand journal of geology and geophysics 6*: 388–394.

Bull, C.; Irving, E. 1960: The palaeomagnetism of some hypabyssal intrusive rocks from south Victoria Land, Antarctica. *Geophysical journal of the Royal Astronomical Society 3*: 211–224.

Bull, C.; Irving, E.; Willis, I. 1962: Further palaeomagnetic results from south Victoria Land, Antarctica. *Geophysical journal of the Royal Astronomical Society 6*: 320–336.

Burmester, R. F.; Sheriff, S. D. 1980: Paleomagnetism of the Dufek Intrusion, Pensacola Mountains, Antarctica. *Antarctic journal of the United States 15*: 43–44.

Dalziel, I. W. D.; Garrett, S. W.; Grunow, A. M.; Pankhurst, R. J.; Storey, B. C.; Vennum, W. R. 1987: The Ellsworth-Whitmore crustal block: its role in the tectonic evolution of West Antarctica. Pp. 173–182 *in*: McKenzie, G. D. *ed.* Gondwana six: structure, tectonics, and geophysics. Washington, D.C., American Geophysical Union. 323 p.

Elliot, D. H.; Fleck, R. J.; Sutter, J. F. 1985: Potassium-argon age determinations of Ferrar Group rocks, central Transantarctic Mountains. *Antarctic research series 36, paper 10* (American Geophysical Union): 197–224.

Fisher, R. A. 1953: Dispersion on a sphere. *Proceedings of the Royal Society of London series A 217*: 295–305.

Ford, A. B.; Kistler, R. W. 1980: K-Ar age, composition, and origin of Mesozoic mafic rocks related to Ferrar Group, Pensacola Mountains, Antarctica. *New Zealand journal of geology and geophysics 23*: 371–390.

Ford, A. B.; Reynolds, R. L.; Huie, C.; Boyer, S. J. 1980: Geological investigation of the Dufek intrusion, Pensacola Mountains, Antarctica. *Antarctic journal of the United States 14*: 9–11.

Funaki, M. 1984: Paleomagnetic investigation of Ferrar Dolerite in the McMurdo Sound region, Antarctica. *Journal of geomagnetism and geoelectricity 36*: 529–563.

Grunow, A. M.; Dalziel, I. W. D.; Kent, D. V. 1987: Ellsworth Mountains–Whitmore Mountains crustal block, western Antarctica: new palaeomagnetic results and their tectonic significance. Pp. 161–171 *in*: McKenzie, G. D. *ed.* Gondwana six: structure, tectonics, and geophysics. Washington, D.C., American Geophysical Union. 323 p.

Gunn, B. M.; Warren, G. 1962: Geology of Victoria Land between the Mawson and Mulock Glaciers, Antarctica. *New Zealand Geological Survey bulletin 71*: 157 p.

Kellogg, K. S. 1980: Paleomagnetic evidence for oroclinal bending of the southern Antarctic Peninsula. *Geological Society of America bulletin 91*: 414–420.

Kellogg, K. S.; Reynolds, R. L. 1978: Paleomagnetic results from the Lassiter Coast, Antarctica, and a test for oroclinal bending of the Antarctic Peninsula. *Journal of geophysical research 83*: 2293–2299.

Kyle, P. R.; Elliot, D.; Sutter, J. F. 1981: Jurassic Ferrar Supergroup tholeiites from the Transantarctic Mountains, Antarctica, and their relationship to the initial fragmentation of Gondwana. Pp. 283–287 *in*: Cresswell, M. M.; Vella, P. *ed.* Gondwana five. Rotterdam, A. A. Balkema. 339 p.

Long, W. E. 1965: Stratigraphy of the Ohio Range, Antarctica. *In*: Hadley, J. B. *ed.* Geology and paleontology of the Antarctic. *American Geophysical Union Antarctica research series 6*: 71–116.

Løvlie, R. 1979: Mesozoic palaeomagnetism in Vestfjella, Dronning Maud Land, East Antarctica. *Geophysical journal of the Royal Astronomical Society 59*: 529–537.

McCartan, L.; Bradshaw, M. A.; Kellogg, K. S. 1986: Reconnaissance geologic map of the Ohio Range, Transantarctic Mountains, Antarctica. *U.S. Geological Survey map A-13*. Scale 1:50 000.

McElhinny, M. W. 1973: Palaeomagnetism and plate tectonics. Cambridge University Press. 358 p.

McIntosh, W. C.; Kyle, P. R.; Cherry, E. M.; Noltimier, H. C. 1982: Paleomagnetic results from the Kirkpatrick Basalt Group, Victoria Land. *Antarctic journal of the United States 17*: 20–22.

Manzoni, M.; Nanni, T. 1977: Paleomagnetism of Ordovician lamprophyres from Taylor Valley, Dry Valleys, Antarctica. *Pure and applied geophysics 115:* 961–977.

Mirsky, A. 1969: Geology of the Ohio Range – Liv Glacier area. *American Geographical Society, Antarctic map folio series, folio 12*. Sheet 17: scale 1:1 000 000.

Ostrander, J. H. 1971: Paleomagnetic investigations of the Queen Alexander Range, Antarctica. *Antarctic journal of the United States 6*: 183–185.

Treves, S. B. 1965: Igneous and metamorphic rocks of the Ohio Range, Horlick Mountains, Antarctica. *In*: Hadley, J. B. *ed.* Geology and paleontology of the Antarctic. *American Geophysical Union Antarctic research series 6*: 117–125.

Turnbull, G. 1959: Some palaeomagnetic measurements in Antarctica. *Arctic 12*: 151–157.

Zijderveld, J. D. A. 1967: A. C. demagnetization of rocks: analysis of results. *In*: Collinson, D. W.; Creer, K. M.; Runcorn, S. K. *ed.* Methods in paleomagnetism. Amsterdam, Elsevier. 354 p.

———1968: Natural remanent magnetism of some intrusive rocks from the Sør Rondane Mountains, Queen Maud Land, Antarctica. *Journal of geophysical research 73*: 3773–3785.